Dual-Representation Mondrian Conformal Assignment of Pressure-Transient GeoTypes: A Reproduction and Extension for Fractured-Reservoir Flow

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***Abstract*—Fractured reservoirs exhibit recurring fluid-flow behaviours hidden in the shape of their pressure-transient responses. A recent study clusters geologically consistent discrete-fracture-network (DFN) ensembles into flow-behaviour classes and attributes each class to fracture-network properties. We first reproduce that pipeline, DFN pressure transients, Bourdet-derivative features, dynamic-time-warping (DTW) k-medoids clustering into a GeoType catalogue, and random-forest / Tree-SHAP attribution, on the original open 4TU corpus and on analytic dual-porosity families, recovering clean clusters (silhouette up to 0.86 on the real transients, higher than on the analytic families) and the finding that fracture intensity $(\log I)$ is the top attributed control. We then extend it in two ways. First, we make the assignment *coverage-controlled*, which the base random-forest attribution is not, with a class-conditional (Mondrian) split-conformal assigner on a DTW nonconformity score that emits per-class p-values, a prediction set, and an out-of-catalogue flag (empirical coverage 0.85–0.87 at target, out-of-catalogue rate 0.03–0.15, mean set size 1.5–1.7). Second, and the methodological contribution, we introduce a *dual-representation Mondrian conformal* assigner that conformalizes jointly across two nonconformity scores, the DTW shape distance and a physics-descriptor implausibility from the attribution forest, and accepts a GeoType only in the per-class conjunction of the two prediction sets. This excludes curves that are shape-close to the wrong physics, which single-score conformal accepts: on the rich-method cases the dual conjunction tightens the mean set size below one and flags right-shape-wrong-physics curves at an honest cost of coverage (e.g. $0.85 \to 0.73$, $0.87 \to 0.75$). The shape machinery is signal-agnostic and is exercised on hydrogeology pumping-test field data as well as reservoir transients. The contribution is a faithful reproduction plus a coverage-controlled assignment that fuses shape space and physics-descriptor space, with every number backed by a committed deterministic artifact.**

***Index Terms*—fractured reservoirs, pressure-transient analysis, discrete fracture networks, dynamic time warping, k-medoids, conformal prediction, Mondrian, random forests, SHAP, reproducibility, geothermal, hydrogeology.**

I. Introduction

THE pressure-transient response of a well is a fingerprint of the reservoir's flow structure: its shape, and the shape of its Bourdet derivative [3], distinguishes homogeneous from dual-porosity behaviour and localizes the interporosity transition [2]. In fractured reservoirs the mapping from geology to transient is many-to-one and uncertain, so a natural question is whether the space of transients organizes into a small catalogue of recurring flow behaviours, a "geological DNA", and if so, which fracture-network properties control membership. A recent study answers both by clustering geologically consistent discrete-fracture-network (DFN) ensembles into flow-behaviour classes and attributing them [1].

This paper reproduces that pipeline and extends it with a coverage-controlled assignment. Reproduction matters here for two reasons: the original workflow depends on a GPL-licensed code base we deliberately do not vendor, and a clustering-plus-attribution result is only trustworthy if an independent re-implementation on the same open data recovers it. We re-implement the shape machinery from permissively licensed components (DTW [5], k-medoids, silhouette selection [6], random forests [7], TreeSHAP [8]) as the open package `pygeotypes`, run it on the original 4TU corpus [11] and on analytic dual-porosity families, and confirm the clusters and the top attributed control.

The extension addresses a gap in how such catalogues are *used*. Random-forest attribution ranks controls but gives no guarantee on a new curve's class, and a curve can be shape-close to the wrong physics. We therefore add conformal prediction [9], [10]: first a class-conditional (Mondrian) split-conformal assigner on the shape distance, then a dual-representation variant that conformalizes jointly over shape and physics-descriptor scores. Our contributions are: (i) an independent, permissively licensed reproduction of the DFN-transient GeoType pipeline that recovers its clusters and fracture-intensity attribution; (ii) a coverage-controlled Mondrian conformal GeoType assigner; (iii) a dual-representation Mondrian conformal assigner that excludes right-shape-wrong-physics curves, with the coverage/tightness trade-off reported honestly.

II. Reproduced pipeline

A GeoType catalogue is built in four stages. *Transients and features.* Each DFN ensemble (or analytic study) yields pressure transients; from each we compute the log-time pressure and its Bourdet derivative [3], and for analytic dual-porosity

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Preprint, 2026. Code, committed derived artifacts, and the interactive workbench are released under MIT at https://github.com/fsantibanezleal/CAOS_RES_Pulso; the reusable shape machinery is the Apache-2.0 package `pygeotypes` (https://github.com/fsantibanezleal/CAOS_GeoTypes). No GPL workflow code or raw licensed data is redistributed.

references the Warren–Root response [2] is evaluated by Stehfest numerical Laplace inversion [4]. *Clustering.* Curves are compared by dynamic time warping [5] and grouped by k-medoids; the catalogue size K is chosen by a silhouette sweep [6], and each GeoType is represented by its medoid curve and a membership. *Attribution.* A random forest [7] predicts the GeoType from physical descriptors of the network, accuracy-gated (attribution is withheld when the forest cannot predict the labels), and TreeSHAP [8] ranks the controlling descriptors. *Assignment.* A new curve is classified against the baked medoids.

On the original open corpus the reproduction recovers the reported structure: the real 4TU transients cluster more cleanly than the analytic families (silhouette up to 0.86 versus ≈ 0.5–0.6 analytic), and fracture intensity ($\log I$) is the top attributed control (largest mean absolute SHAP), partially reproducing the paper's fracture-intensity finding. On the analytic families the interporosity flow coefficient $\log_{10}\lambda$ dominates the assignment sensitivity, exactly as the physics predicts, since λ sets the transition timing that separates the dual-porosity behaviours.

III. Coverage-controlled assignment

A. Mondrian split-conformal on the shape score

Random-forest attribution ranks controls but offers no guarantee that a new curve is assigned to the right GeoType. We add a class-conditional (Mondrian) split-conformal assigner [9], [10]. The nonconformity score of a curve x for GeoType g is its DTW distance to the medoid, $s_{\text{shape}}(x, g)$. Calibrated per class on a disjoint slice, it yields a p-value $p_{\text{shape}}(x, g)$ and a prediction set $\{g : p_{\text{shape}}(x, g) > \alpha\}$; a curve matching no class is flagged out-of-catalogue. Across the rich-method cases the empirical marginal coverage is 0.85–0.87 at the target level, the out-of-catalogue rate is 0.03–0.15, and the mean set size is 1.5–1.7: a coverage guarantee the base attribution cannot give.

B. Dual-representation Mondrian conformal (contribution)

A single score cannot express that a curve may be shape-close to the wrong physics: a transient that matches a medoid by DTW shape yet whose physical descriptors are atypical for that GeoType. We conformalize jointly across two representations. In addition to the shape score, define a descriptor score

$$s_{\text{desc}}(x, g) = 1 - \mathrm{RF}\big(g \mid \text{descriptors}(x)\big), \tag{1}$$

the implausibility of x's physical descriptors under GeoType g from the attribution forest. Both are Mondrian-calibrated, giving p_{shape} and p_{desc}, and the dual prediction set is the per-class conjunction

$$\Gamma^{\alpha}(x) = \big\{\, g : p_{\text{shape}}(x, g) > \alpha \;\wedge\; p_{\text{desc}}(x, g) > \alpha \,\big\}. \tag{2}$$

A GeoType is accepted only if the curve is both shape-consistent and descriptor-consistent; a shape match with implausible physics is excluded. This is beyond single-score conformal and beyond random-forest/SHAP attribution, which has no coverage control: a coverage-controlled assignment that fuses shape space and physics-descriptor space.

TABLE I
Dual-representation vs. shape-only Mondrian conformal. The dual conjunction trades marginal coverage for tighter, physics-consistent sets and excludes right-shape-wrong-physics curves.

Case	shape-only coverage	dual coverage (mean set)
Rich-method A	0.85	0.73 (0.78)
Rich-method B	0.87	0.75 (0.89)

Table I reports the honest trade-off on two rich-method cases. The dual conjunction tightens the mean set size below one and flags right-shape-wrong-physics curves that shape-only conformal accepts, at the cost of some marginal coverage ($0.85 \to 0.73$, $0.87 \to 0.75$). When descriptors are absent or degenerate (some real cases), the dual layer degrades gracefully to shape-only and reports that it did.

IV. Generalization and honest scope

The shape machinery is signal-agnostic: any response signal whose shape reflects the underlying physical system, hydrogeology pumping tests, geothermal and thermal-response tests, can be catalogued and conformally assigned by the same code. We exercise it on real hydrogeology pumping-test field data (the Horkheim and Lauswiesen sites) alongside the reservoir transients, all through one pipeline.

Scope is reported plainly. This is a reproduction plus extension of a published pipeline, not a wholly new method for the base clustering and attribution, which are credited to the reproduced work. The DFN pressure-transient simulation on the full open-source flow engine is an offline dependency whose end-to-end integration is a later phase; the results here rest on the paper's own committed corpus, analytic dual-porosity families, and DFN ensembles, and the artifacts that lack a simulated transient carry an explicit pending flag rather than a fabricated curve. The dual-conformal coverage is a genuine trade: it buys physics-consistency and tighter sets at the price of marginal coverage, and both sides are shown. Every number is a committed, seed-deterministic artifact.

V. Conclusion

We independently reproduced, from permissively licensed components, a DFN-transient GeoType pipeline for fractured-reservoir flow, recovering its clusters (silhouette up to 0.86) and its fracture-intensity attribution, and extended it with a coverage-controlled Mondrian conformal assigner and, as the methodological contribution, a dual-representation Mondrian conformal assigner that fuses shape and physics-descriptor evidence to exclude right-shape-wrong-physics assignments. The result is a reproducible, coverage-controlled way to place a new pressure transient in a catalogue of flow behaviours and to say, with a guarantee, when it belongs to none.

Data and code availability

Code, the committed derived artifacts, and the workbench are at https://github.com/fsantibanezleal/CAOS_RES_

Pulso (MIT); the reusable shape machinery is `pygeotypes` (Apache-2.0). The reference corpus is the open 4TU dataset of the reproduced study [11] (GPL-3.0, kept in a local vault, never committed); GPL workflow code is not vendored. Only compact derived artifacts are committed.

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